Leveraging Large Language Models to Generate Original and Cogent Arguments on the Connectionist Eliminativism of Belief*

Sangmin Seo

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Logic and Philosophy of Science
Munich Center for Mathematical Philosophy
Fakultät für Philosophie, Wissenschaftstheorie und Religionswissenschaft
Ludwig-Maximilians-Universität München

Supervisor: Dr. Ignacio Ojea Quintana
Second Reader: Dr. David Colaco

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Abstract

Large language models (LLMs) excel in many tasks, but their effectiveness in generating original, coherent philosophical arguments remains uncertain. In this paper, I focus on arguments related to connectionist eliminativism of belief and employ iterative critique-revision loops on LLM-generated argument outlines to enhance their originality and cogency. Using the enhanced outlines, I generate full argument texts, applying either instructional prompts or fine-tuning techniques to promote cogency and originality. Results show that as the number of critique-revision loops increased, the semantic textual similarity between the iterated outlines and existing relevant argument outlines decreased, indicating greater originality. However, human evaluators found the iterated outlines less cogent than the non-enhanced ones. Additionally, when generating full argument texts from outlines, the reasoning model (o1) produced more cogent and original arguments than GPT-4o and fine-tuned GPT-4o under the same instructional conditions.

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1 Introduction

Despite the impressive achievements of large language models (LLMs) across various natural language processing tasks, it remains uncertain whether they can generate original and cogent arguments in philosophy. By "cogent" argument, I refer to the argument whose reasons are *plausible* and the reasons genuinely *support* the claim being made. I define plausible reasons as those that are likely to be true, reasons supporting a claim as those that help to show the claim to be true, and *original* reasons as those that seem not the same as any other arguments' reasons, in line with the Cambridge Dictionary's definition.

While recent LLMs can produce arguments that are not original and may include some unacceptable reasons, generating truly original and cogent arguments appears to be more challenging. For instance, with appropriate prompt engineering, LLM-based chatbots like ChatGPT can generate arguments that support a claim. An academic example is DigiDan, a GPT-3 model fine-tuned on Daniel Dennett's works, which produces responses that resemble Dennett's answers to philosophical questions. (Schwitzgebel, Schwitzgebel, & Strasser, 2024) Although this model could generate arguments somewhat similar to Dennett's, it was not designed to produce original philosophical arguments.

Crafting an original, plausible, and supporting argument is essential for writing a strong paper not only in analytic philosophy but also in many other academic disciplines. Investigating the potential for automatically generating original and cogent arguments is crucial, as it could significantly impact the practice of academic fields like philosophy, where a high level of natural language understanding is required.

To make the research problem more tractable, I focus on one specific controversy in philosophy: the connectionist eliminativism of belief. Eliminative materialism, concerning mental states, is the philosophical position that concepts like belief, desire, or even consciousness—often referred to as folk-psychological concepts—are not real and will be eliminated from scientific discussions of the mind, much like how the Phlogiston and Aether theories were discarded (Ramsey, 2022). Connectionism explains mental processes through neural connections, their weights, and neural activations (C. Buckner & Garson, 2019; Waskan, 2024) Connectionist eliminativism of belief is the thesis that if the connectionist networks they describe accurately model cognitive systems, then eliminativism regarding propositional attitudes, such as beliefs, is justified.

Focusing on this specific topic makes it easier to evaluate the originality and cogency of generated arguments. It is because the number of literature directly related to it is relatively small (around 30), and there is room for new arguments that reflect the implications of deep learning models, such as LLMs, since connectionist

eliminativism was primarily explored in the 1990s (Piotrowski, 2024). Accordingly, this research addresses the following research question.

Can LLMs generate original and cogent philosophical arguments based on specific claims in the debate surrounding connectionist eliminativism of belief?

Examining connectionist eliminativism of belief is crucial because connectionist approaches are increasingly influential in fields such as neuroscience, psychology, and computer science. Moreover, the potential elimination of belief as a concept has significant implications for epistemology and the philosophy of mind. Given the relevance of this philosophical issue, I propose the following hypothesis in response to the main research question.

Properly configured LLMs, when combined with methods that enhance originality and cogency, can generate philosophical arguments on the connectionist eliminativism of belief that are significantly more original and cogent, comparable to those produced by prompting a single LLM.

If successful, this approach could enrich philosophical discussions on topics like connectionist eliminativism of belief and highlight the potential of LLMs in contributing to philosophical debates. On the other hand, if unsuccessful, it would underscore the inherent challenges of philosophical argumentation or the limitations of LLMs in this domain.

1.1 The Debate Over Connectionist Eliminativism of Belief

Let us examine the debate on connectionist eliminativism of belief before generating arguments about it.

Ramsey, Stich, and Garon (1995) (hereafter RS&G) argue that if the connectionist networks they describe accurately model cognitive systems, then eliminativism regarding propositional attitudes, such as beliefs, is justified. This conclusion arises from the claim that folk psychology's portrayal of propositional attitudes is incorrect—specifically, that propositional attitudes lack propositional modularity if connectionism is valid.

For the summary on propositional modularity and connectionist models, I used OpenAI's o1 model (2024-12-17), drawing from several paragraphs in the sections "3 Propositional Attitudes and Common-Sense Psychology" and "4 A Family of Connectionist Hypotheses" in Ramsey et al. (1995). The following summaries were used as reference information to assist in generating argument outlines and collecting human evaluations on those outlines (see and subsection 4.1).

> Propositional modularity refers to the concept that propositional attitudes, such as beliefs and desires, are functionally discrete, semantically interpretable, and play a causal role in behavior. In common-sense psychology, these attitudes are seen as distinct states that can independently influence actions. For instance, Alice's decision to go to her office could be driven by her desire to send emails or to talk to her research assistant. Each desire represents a semantically distinct state, and either could be causally active in prompting her behavior. This illustrates how propositional attitudes are modular, with each having the potential to

independently affect outcomes based on their causal roles. (Ramsey, Stich, and Garon, 1995))

Connectionist models are networks composed of simple, parallel computing units that process information by propagating activation values through connections with varying strengths, or weights. These models transform input data into output through a series of interconnected units, including hidden units that facilitate complex computations without explicit instructions. The network's knowledge is encoded in the connection strengths and biases, which are adjusted through learning procedures. These procedures often involve training with input/output pairs, allowing the network to autonomously tune its weights and biases to perform specific computations. This self-programming capability is a key feature of connectionist models, enabling them to adapt and learn from data. (Ramsey et al., 1995; Smolensky, 1995))

A belief has propositional modularity if and only if the belief is functionally discrete, semantically interpretable, and possess a causal role. RS&G seem to allow for the possibility of semantic interpretability within connectionist models (as noted by Macdonald (1995)). However, they argue that no connectionist model demonstrates a discrete component representing propositions, and the information in connectionist models does not fulfill functionally discrete causal roles, as information is distributed and sub-symbolic within such networks.

In response to RS&G, Clark (1995) argues that connectionism does not necessarily lead to eliminativism regarding belief. Using cluster analysis of an actual connectionist network, $NETtalk$, he demonstrates that sufficiently complex connectionist models can exhibit propositional modularity.

Cluster analysis involves averaging the activation vectors of hidden units for each output and grouping similar averaged vectors. For example, in the text-to-speech model $NETtalk$, cluster analysis can group activation patterns producing vowels into one category and those producing consonants into another. According to Clark, a transient belief that P corresponds to the activation patterns of hidden units clustered as P, while a long-standing belief that P corresponds to the disposition to produce these activation patterns.

Clark suggests that under cluster analysis, connectionist networks can contain semantic and discrete entities that qualify as beliefs. Moreover, beliefs in a network can play discrete causal roles. It is because a specific activation pattern, as opposed to an alternative pattern, can produce a distinct output, even when two different long-standing beliefs within the network result in the same output. This phenomenon is referred to as the "equipotency claim."

Smolensky (1995a) also challenges the conclusions of RS&G, arguing that connectionism does not necessarily lead to eliminativism. He introduces the concepts of C-belief and L-belief in relation to the models $RSGnet_0$. $RSGnet$ refers to the example model used by RS&G to illustrate connectionist networks, while $RSGnet_0$ is a simplified version of $RSGnet$ presented by Smolensky.

Through weight analysis, Smolensky identifies C-beliefs in $RSGnet_0$, while learning analysis reveals the presence of L-beliefs. For an input vector p representing a proposition P, if the weight vector w lies in the positive (or negative) half-space of

p, then $RSGnet_0$ holds the C-belief that P is true (or false). Given a vector p^*, where $p \cdot p^* = +1$ and p^* is orthogonal to other training propositions, the "L-belief that P has truth value tr(P)" is defined as tr(P)p^*.

Both C-beliefs and L-beliefs exhibit semantic interpretability, consistent with Smolensky's semantic level principle (Smolensky, 1995b), which states that distributed regions in a network define semantic entities. They also demonstrate functional discreteness: C-beliefs can be distinguished in vector space, and L-beliefs are expressed discretely for each proposition. (However, as noted in Macdonald (1995) Smolensky does not fully clarify the basis for L-beliefs' functional discreteness.)

Despite this, C-beliefs and L-beliefs do not fulfill the requirement of playing discrete causal roles, as the equipotency claim appears implausible for both. While these beliefs are not fully propositionally modular, they are not eliminated.

Additionally, Smolensky argues that $RSGnet_0$ is an equally valid model as $RSGnet$ for illustrating the properties of connectionist networks. He further claims that C-beliefs and L-beliefs also exist in $RSGnet$, though the theoretical support for this assertion is incomplete. Smolensky conjectures that future connectionist models, more complex than $RSGnet_0$ and $RSGnet$, might incorporate explanatory notions such as C-beliefs and L-beliefs.

Stich and Warfield (1995) (hereafter S&W) argue that connectionism does not necessarily entail eliminativism, critically reviewing the works of Ramsey et al. (1995), Clark (1995), and Smolensky (1995a). S&W decompose RS&G's connectionist eliminativism thesis into two key claims:

1. If connectionism is correct, then there are no propositionally modular beliefs.
2. If there are no propositionally modular beliefs, then eliminativism regarding belief is justified.

S&W assert that Clark refutes (1) while tacitly accepting (2), whereas Smolensky challenges (2) but seems to partially accept (1). S&W themselves critique (2) from a perspective distinct from Smolensky's while ultimately accepting (1), offering criticism of both Clark and Smolensky.

S&W argue that the $NETtalk$ models referenced in Clark (1995) do not reliably exhibit a common clustering profile. Different $NETtalk$ networks with variations in the number of units and connections can be built. In fact, $NETtalk$ with additional hidden units process the same input in distinct ways. As a result, the activation patterns of hidden units in connectionist networks like $NETtalk$ are unlikely to be semantically interpretable or functionally discrete.

Moreover, S&W contend that the concept of propositional modularity employed in Clark (1995) is overly broad, to the extent that any input-output device with internal patterns could be said to possess propositionally modular beliefs.

Clark supports the equipotency claim by appealing to the recurrence of output as input (which is one of the beliefs yielding the same result). However, S&W argue that this approach—requiring a discrete intermediate step for inference—is incompatible with the parallel information processing characteristic of connectionist networks. Additionally, recurrence fails to account for more complex forms of inference, as demonstrated by an example provided by S&W.

In relation to Smolensky (1995a), S&W highlight the lack of clarity regarding whether more complex connectionist networks—beyond the simplified $RSGnet_0$ and $RSGnet$—can contain semantic entities such as C-beliefs and L-beliefs. The theoretical foundations supporting the existence of such entities in more complex networks remain underdeveloped, leaving the applicability of Smolensky's arguments uncertain.

S&W challenge the claim (2) by arguing that even if beliefs lack propositional modularity, they can still exist, provided that one of two problematic premises is not assumed. This is because there is a logical gap between rejecting folk psychology and adopting eliminativism.

One premise that attempts to bridge this gap is the description theory of reference, which suggests that beliefs do not refer to anything if the theory describing them is proven false. However, the description theory has been widely criticized, particularly in light of the causal/historical theory of reference, which dominates contemporary discussions on the topic (e.g., Kripke (1980)).

Another premise that could fill this logical gap is the claim that the properties of propositional modularity are constitutive of (or necessary for) belief. However, this premise relies on the idea that the statement "beliefs are propositionally modular" is analytic. This notion of analyticity has been subjected to influential critiques, such as those presented by Quine (1953).

Therefore, to defend (2), it is necessary to refute the criticisms of these underlying premises and establish their validity. Without addressing these foundational issues, the argument for eliminativism remains unconvincing.

1.2 Argument Frameworks Based on the Debate

The authors recognize a critical issue: the implications of connectionism for mental states such as belief. This problem is particularly significant because connectionist models are widely applied in both academic fields like neuroscience and practical domains such as LLMs. Consequently, their explanations of mental phenomena need further development. In addressing this problem, the authors share a central question: Does connectionism imply the eliminativism of propositional attitudes like belief? Following S&W's observation regarding RS&G's connectionist eliminativism, this question can be divided into two components:

1. Does connectionism imply that propositional modularity is incorrect?

2. If propositional modularity is incorrect, does this justify eliminativism of belief?

This paper focuses on the first question to generate arguments. Using a question in the prompt can enhance originality by encouraging the generation of novel claims. However, to maintain consistency and facilitate comparison, reasons supporting a fixed claim are generated. Two opposing claims respond to the first question:

(1) Propositional modularity is incompatible with connectionist models of cognitive systems.

(2) Functionally discrete beliefs, playing a causal role, can be compatible with connectionist models of cognitive systems.

RS&G and Stich support claim (1), and Smolensky assumes claim (1). In con-

trast, Clark supports claim (2). Additionally, RS&G and Clark each present sections explicitly supporting or opposing these claims, enabling direct comparisons based on these outlines. The claim associated with the second question—'If propositional modularity is incorrect, eliminativism of belief is justified'—is directly criticized by Smolensky and S&W. This claim, while relevant to the primary claims, will be treated as a baseline for comparison, as the reasons supporting it differ substantially from those related to claims (1) or (2).

1.3 Foundational Concepts for Generating Texts through Large Language Models

How can cogent and original arguments be generated using LLM frameworks? This subsection introduces key concepts underlying LLM techniques for text generation.

According to Jurafsky and Martin (2024), the distributional hypothesis posits that word meanings are determined by their contextual distributions—words appearing in similar contexts share similar meanings. Vector embeddings represent words numerically within a vector space, enabling natural language processing tasks such as text generation and similarity computations. For instance, semantic textual similarity (STS) is calculated using the cosine similarity between vector embeddings of two texts: $\text{STS} = \cos\theta = \frac{a \cdot b}{|a||b|}$, where a and b are embeddings for texts A and B, respectively.

The language models discussed in this paper are based on deep learning, which enhances the capabilities of artificial neural networks by introducing multiple hidden layers between input and output layers (C. J. Buckner, 2023; C. Buckner & Garson, 2019) (Goodfellow, Bengio, & Courville, 2016, p.163). Deep learning has demonstrated high performance in tasks requiring intuitive abilities. For example, despite the immense number of possible tracks in Go (10^{360}), deep learning models such as AlphaGo have surpassed human champions by exploring fewer moves more effectively. Beyond board games, deep learning models have achieved remarkable success in identifying objects in images, generating realistic videos and music, and more (C. J. Buckner, 2023).

The transformer is a deep learning architecture that forms the foundation of large language models. It excels at capturing contextual information to generate coherent text, even when important elements within the context are far apart. The query, key, and value (QKV) representations within the transformer are central to its ability to capture context.

- Query vectors represent the tokens whose contextual meaning we aim to understand.
- Key vectors represent other tokens that influence the meaning of the query.
- Value vectors represent the contextualized meaning of the query tokens, shaped by the keys. (C. J. Buckner, 2023)

When matrix Q contains query vectors, K contains key vectors, and V contains value vectors, the output of scaled dot-product attention is given by:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where d_k is the dimension of the query and key vectors. The scaled dot-product attention mechanism processes the query, its context, and its contextual meaning simultaneously.

Multi-head attention further enhances this mechanism by projecting Q, K, and V multiple times, applying scaled dot-product attention to each projection in parallel, and then concatenating the results. (Vaswani et al., 2017)

Transformer-based models, pre-trained on large-scale corpora such as webpages, have demonstrated state-of-the-art performance in natural language processing (NLP) tasks, including text generation. They surpassed task-specific models when fine-tuned for specific tasks. (Radford, Narasimhan, Salimans, & Sutskever, 2018; Devlin, Chang, Lee, & Toutanova, 2019)

Larger models can excel on many NLP tasks through zero-shot prompts (solving tasks without prior examples) and k-shot prompts (providing contextual examples), achieving strong results even without fine-tuning. (Radford et al., 2019; Brown et al., 2020)

Fine-tuning pre-trained large language models using supervised or reinforcement learning further enhances their alignment with user intent. (Achiam et al., 2023; Wang et al., 2023; Ouyang et al., 2022) For instance, GPT-4 generates synthetic data, identifies and removes hallucinations from its outputs, and uses both flawed and corrected responses for reinforcement learning, significantly improving factual accuracy. (Achiam et al., 2023, pp.64-65)

Additionally, self-instruction is a powerful fine-tuning approach. Wang et al. (2023) synthesized instruction prompts from seed examples, and a GPT-3 model fine-tuned on this synthetic data outperformed InstructGPT, which had been fine-tuned using human feedback.

Reasoning models, such as OpenAI's o1-preview and o1, which are used in this paper, are LLMs fine-tuned through reinforcement learning to generate chain-of-thought reasoning before providing responses (OpenAI et al., 2024). Chain-of-thought reasoning consists of "a series of intermediate reasoning steps." It is well-established that prompting an LLM to produce chain-of-thought reasoning, such as by providing examples with detailed reasoning steps, significantly improves problem-solving performance (Wei et al., 2022). Furthermore, models supervised on intermediate chain-of-thought processes demonstrate substantially better performance than those supervised solely on final outcomes (Lightman et al., 2023). Reasoning models, in particular, excel in STEM fields (OpenAI et al., 2024), which demand complex and logical reasoning.

To contextualize these concepts within the scope of this paper, let us review them. Vector representations, or embeddings, serve as fundamental elements for processing the meanings of natural language objects, exemplifying the distributional hypothesis of meaning. Deep learning has NLP tasks that require intuitive reasoning. The Transformer architecture plays a crucial role in capturing contextual word meanings, even over long sequences. Pre-training large Transformer-based models on vast text corpora allows them to generate coherent and contextually appropriate text. Addi-

tionally, fine-tuning LLMs with outputs aligned to intended or desirable responses further improves their performance on general tasks. Notably, LLMs exhibit strong performance on reasoning-intensive tasks when fine-tuned to utilize chain-of-thought reasoning.

Due to recent technical advancements, LLMs now outperform humans in many tasks and even surpass experts in certain domains when provided with well-crafted prompts. This capability is enhanced through a technique known as prompt engineering, which improves model responses by carefully designing prompts (OpenAI, 2024). Few-shot learning and chain-of-thought prompting are notable forms of prompt engineering. I anticipate that advanced LLMs, such as GPT-4o and o1, can reconstruct argument texts into structured outlines and generate high-quality argument outlines and texts when guided by properly engineered prompts.

2 Enhancing Originality and Cogency of Argument Outlines

2.1 Background and Method

DigiDan (Schwitzgebel et al., 2024) presents concise arguments modeled after the style of philosopher Daniel Dennett. While many of these arguments achieve a certain degree of cogency, they often lack originality, largely reiterating Dennett's established ideas, as Schwitzgebel et al. (2024) intended. With present LLMs like GPT-4o and Gemini 1.5 series, generating cogent arguments is relatively straightforward with minimal prompting. However, achieving both originality and a distinct philosophical style presents a more complex challenge.

One potential method to foster originality is generating numerous arguments and identifying the original ones from the pool. However, this approach is impractical for non-experts in the relevant domain and time-consuming even for seasoned experts. An alternative strategy involves modifying parameters such as temperature and top_p during the argument generation process.

Higher temperature increases the disparity between the probabilities of different options, resulting in a steeper probability distribution. This makes the model more likely to select candidates with higher probabilities. Similarly, raising the top_p value (e.g., from 0.5 to 0.8) broadens the range of next-token probabilities by including less likely options. While these adjustments can lead to more original outcomes, they often compromise the cogency of the generated arguments by introducing less probable or contextually less relevant elements. Therefore, the method enhancing originality and not trading-off cogency is needed.

Conversely, lower temperature and lower top_p values produce more predictable responses, as the model prioritizes the most probable answers based on the context. This is especially useful for factual outputs. These parameters are not available for reasoning models like o1. I used a temperature of 1.0 (default) and top_p of 0.01 for GPT-4o's generations to obtain more factual and reproducible responses, following OpenAI API's recommendation not to modify both values simultaneously.

To enhance the cogency and originality of argument outlines, I employ critique-

and-revision loops. This method is expected to improve not only the plausibility of the arguments, as seen with GPT-4's reduction in generating false information, (Achiam et al., 2023, p.64) but also to strengthen the support and originality of the generated reasons.

The mechanism is designed to improve the cogency and originality of arguments by employing a divided, reflective, and human-like process. It breaks down the task of generating cogent and original arguments into intermediate steps, which include: writing reasons supporting a claim, repeating to critique (or criticized) on the cogency and originality of the outline and revising the outline by reflecting the critiques.

This approach is inspired by examples where dividing the reasoning process, incorporating reflection, and mimicking human-like processes have led to improved outcomes. For instance, the Chain of Thought (CoT) mechanism enhances response quality by prompting a language model with examples of human-like divided reasoning. Wei et al. (2022)

Similarly, AlphaGeometry, which can solve most Olympiad-level geometry problems, employs a cyclical process between a geometric theorem prover and a neural language model. When the theorem prover exhausts all possible deductions and is unable to complete the proof, the language model steps in to construct a new geometric proposition. This proposition is then used by the theorem prover to attempt the proof again. This cycle between the theorem prover and the language model continues until the target theorem is successfully proved. (Trinh, Wu, Le, He, & Luong, 2024) This approach mirrors the human process of integrating intuition with deductive reasoning.

As discussed briefly, GPT-4 improved its response factuality by using a divided, reflective process. It lists hallucinations in a response and revises the response by cycling through the identification and elimination of hallucinations five times, (Achiam et al., 2023, p.64) comparable to how humans revise their work based on feedback.

The process for enhancing cogency and originality through critique-revision iteration involves three key steps:

1. Generating an initial outline: This is achieved through a prompt designed to produce reasons supporting a specific claim, with defined criteria for length, plausibility, support, and originality. One of the two fixed claims discussed in subsection 1.2 is used.

2. Critiquing the outline: LLM criticizes the generated outline, identifying issues related to the criteria provided. This includes assessing whether the reasons are plausible, whether they effectively support the claim, and whether the reasons are original.

3. Revising the outline: A new outline is generated in response to the paired critique and previous outline.

Steps 2 and 3 are iterated k-times to progressively refine cogency and originality. Two prompt variations are used for these steps: one focusing exclusively on originality and another combining originality with cogency. Simpler versions of

the enhancement methods, which align more closely with GPT-4's hallucination reduction method (Achiam et al., 2023, p.64), are also implemented. One group of outlines is enhanced simply, while the other group undergoes full enhancement. For the simple enhancement, the input for the revision step includes the most recent outline and its corresponding critique. In contrast, the full enhancement's revision step incorporates the entire history of outlines and critiques. For example, the second enhancement's revision input includes (1) the first generated outline, (2) the critique of it, (3) the second outline revised based on the first outline and critique, and (4) the critique of the second outline.

But why target argument outlines instead of full arguments? There are several advantages: First, original or unique outlines are more sensitive to textual differences due to their compact and concise nature, compared to longer texts. Secondly, evaluating short texts is more practical and efficient. Outlines serve as a foundation for prior testing before applying originality-enhancement techniques to complete argument texts. Furthermore, despite their brevity, outlines preserve core content, enabling subsequent generation of argument texts based on these foundational outlines.

Preliminary information about propositional modularity and connectionist models, which was generated using OpenAI o1, is provided in the input. (subsection 1.1) The parts "*Information ... role" and "**Information ... models" are added by me. Although the texts are from RS&G, they do not represent a specific stance of RS&G. Instead, they convey concepts accepted by the authors introduced in subsection 1.1.

The prompts for text generations used markdown formatting. For instance, text enclosed in * is italicized, while text enclosed in ** is bolded. The symbol * is used to represent a literal asterisk (*) for reference purposes. In OpenAI models, system messages provide overarching instructions for the model to follow, and user messages represent the inputs provided by users.

2.2 Examples of Prompts and Responses

The following is the prompt used to generate initial outlines (step 1) and unenhanced outlines. The words "**System Message:**" and "**User Message:**" were not in the actual prompts. Certain texts within {} have been omitted for clarity and to avoid repetition.

> **System Message:**
>
> **Task Prompt:** State reasons to support the main claim provided.
>
> Ensure the following *criteria* are met:
>
> **Length:** The reasons should be around 100 words in total, equivalent to the length of an outline for one body section or subsection.
>
> **Plausibility:** The reasons provided should be plausible. For example, do not hallucinate, and do not make up factual information.
>
> **Support:** The reasons must support the given main claim. For example, stay focused on the main claim without deviating into unrelated claims.

Originality: Your reasons should be original and unique. For example, do not reiterate the well-known arguments.

User Message:

Main Claim: *Propositional modularity* is incompatible with connectionist models** of cognitive systems.*

{Information about Propositional Modularity and Connectionist Models}

Reason (1): [Provide the reason supporting the main claim.]

Reason (2): [Provide the reason supporting the main claim.]

Reason (3): [Provide the reason supporting the main claim, if applicable.]

[Provide additional reasons if applicable.]

The following is the prompt used to critique the provided outline based on three criteria. (step 2)

System Message:

Task Prompt: For the provided reasons, construct a critique that identify any points that are implausible, fail to support the main claim, or lack originality. Your critique should be around 50 words. Do *not* critique, challenge, or comment on the position or stance of the main claim under any circumstances.

User Message:

{Main Claim}

{Information about propositional modularity and connectionist models}

Reasons: “” “{Generated reasons are inserted here.}” “”

Task Reminder: {The same task prompt is reminded here.}

Critique on **Reasons**

Implausible Points:

- [Specify any identified implausible point if applicable.]

[List additional implausible points, if applicable.]

Points Not Supporting the Main Claim:

- [Specify any identified point that does not support the main claim if applicable.]

[List additional points not supporting the main claim, if applicable.]

Non-Original Points:

- [Specify any identified non-original point, if applicable.]

[List additional non-original points, if applicable.]

The following is the prompt used in step 3 to revise argument outlines based on critiques.

System Message:

Task Prompt: Revise the last reasons by removing the points identified in all the critique provided. The revised reasons should *support* the main claim.

{The four criteria for length, plausibility, support, and originality are provided.}

User Message:

{Main Claim}

“” “{Each history of generated outlines and critiques is inserted here.}” “”

{Task Reminder}

{The format of reasons is provided.}

One of the generated reasons, along with its critique and the revised reasons based on the critique, is shown below.

Main Claim:

Main Claim: *Propositional modularity* is incompatible with connectionist models** of cognitive systems.*

Generated Reasons:

Reason (1): Connectionist models represent information through distributed networks where mental states overlap and interact continuously. This contrasts with propositional modularity, which requires functionally discrete propositional attitudes that operate independently.

Reason (2): In connectionist systems, semantic content emerges from complex patterns of activation across the network, lacking explicit semantic interpretability. This incompatibility arises because propositional modularity assumes that propositional attitudes are semantically interpretable as distinct entities.

Reason (3): The causal influence in connectionist models stems from the holistic processing of the network, not from individual propositional attitudes. This challenges propositional modularity's claim that discrete propositional attitudes have distinct causal roles in driving behavior.“” “

Critique on the Reasons:

Implausible Points:

- In Reason (2), claiming connectionist models lack explicit semantic interpretability is implausible, as methods exist to interpret neural activations semantically.

- Reason (3) overlooks that some connectionist units can have distinct causal roles, challenging the idea that causal influence stems solely from holistic processing.

Non-Original Points:

- The reasons presented are standard critiques lacking originality, offering common arguments without novel insights.

Reasons Revised based on the Critique:

Reason (1): Connectionist models rely on distributed representations where information is encoded across networks of interconnected units. This makes it difficult to isolate specific propositional attitudes as functionally discrete entities, as required by propositional modularity. The overlap and integration of activation patterns in connectionist systems suggest an incompatibility with the notion of discrete, independent mental states.

Reason (2): In connectionist systems, learning and processing involve continuous interactions among units, leading to holistic changes in the network's state. This interconnectedness means that individual propositional attitudes cannot have distinct causal roles without influencing others, challenging the propositional modularity claim that such attitudes operate independently to drive behavior.

3 Evaluating Originality of Generated Argument Outlines through Semantic Textual Similarity

3.1 Method

How can we measure originality? My approach involves measuring semantic textual similarity (STS) between generated outlines and outlines summarized from existing arguments. A low STS between generated outlines and outlines based on existing relevant texts may suggest uniqueness, novelty, or originality in the generated outlines, although low STS does not necessarily guarantee these qualities.

For evaluation, 11 outlines are generated for each condition:

1. Non-enhanced. ($\times 2$ claims)
2. 10 simple iterations of enhancement focusing on originality. ($\times 2$ claims, $\times 4$ generations)
3. 10 full iterations of enhancement focusing on originality. ($\times 2$, $\times 4$)

4. 10 simple iterations of enhancement focusing on both originality and cogency. (×2, ×4)

5. 10 full iterations of enhancement focusing on both originality and cogency. (×2, ×4)

6. Irrelevant (×2)

For generating argument outlines, GPT 4o (2024-08-06), o1 preview (2024-09-12), and o1 (2024-12-17) are used. These models are among the best for hard, long-form tasks according to the LLM Arena. (Chiang et al., 2024)[1] o1-preview and o1 are reasoning models, with o1 generally outperforming o1-preview while using fewer reasoning tokens.

The two claims on the first question of connectionist eliminativism (subsection 1.2) were used for each group, and enhanced outlines were generated four times to improve the accuracy of the average. The method for generating non-enhanced outlines is the same as the first step of generating enhanced outlines, which involves prompting for original or cogent-and-original outlines without critique-revision cycles. Irrelevant outlines are generated without enhancement based on a claim unrelated to the outlines being compared. The claim on the second question of connectionist eliminativism (subsection 1.2) is used for generating irrelevant outlines compared with the non-enhanced outlines. ("*Although there are no propositionally modular beliefs, eliminativism regarding belief is not justified.*") The claim of the Chinese Room Argument by John Searle ("*Any digital computer does not understand Chinese soley on the basis of implementing the appropriate program for understanding Chinese.*") is used for generating irrelevant outlines compared with outlines constructed from existing texts.

For comparison, non-enhanced outlines and "outlines constructed from existing texts" are used. Non-enhanced outlines are generated using the same method as the baseline non-enhanced outlines. outlines constructed from existing texts are created by reconstructing sections from existing literature into outlines that follow the same structure as the generated ones. Relevant sections are identified by searching for keywords like "propositional modularity" and "discrete and causal." A total of 51 relevant sections were identified from 245 sections in the literature on connectionist eliminativism, including many sections in Ramsey et al. (1995); Clark (1995); Smolensky (1995a); Stich and Warfield (1995), introduced in subsection 1.1.

Each outline is embedded for STS using Google's text-embedding-005 model (Refer to subsection 1.3 for information about embedding and STS). For each condition, the average STS is measured in two ways: between [Enhanced Outline, Non-Enhanced Outline] pairs, and between

[1] https://lmarena.ai/?leaderboard

[Enhanced Outline, Outline Constructed from Existing Text] pairs. For baseline comparisons, STS is also measured between [Non-Enhanced Outline, Non-Enhanced Outline] pairs and [Non-Enhanced Outline, Outline Constructed from Existing Text] pairs. Additionally, [Irrelevant Outline, Non-Enhanced Outline] pairs and [Irrelevant Outline, Outline Constructed from Existing Text] pairs are used for the second baseline.

To detect trends, the Mann-Kendall (MK) trend test is applied to each list of STS scores. The description of the MK Test follows Laboratory (2025). The MK Test identifies monotonic upward, downward, or no trends, without requiring the data to be normally distributed or linear. The null hypothesis (H_0) posits no monotonic trend, while the alternative hypothesis (H_a) suggests the presence of a monotonic trend. The MK test follows these steps:

1. Data is ordered by time. In this research, STS results measured sequentially from outlines without enhancement (x_1) through outlines enhanced by one cycle (x_2) to outlines enhanced through 10 cycles (x_{11}) are ordered.

2. All possible differences between observations are calculated as $x_j - x_k$ for $j > k$. For instance, $x_2 - x_1$, ..., $x_7 - x_3$, and so on.

3. The indicator function is applied: $sgn(x_j - x_k) = \begin{cases} 1 & \text{if } x_j - x_k > 0 \\ 0 & \text{if } x_j - x_k = 0 \\ -1 & \text{if } x_j - x_k < 0 \end{cases}$

4. The test statistic is computed:

$$S = \sum_{k-1}^{n-1} \sum_{j-k+1}^{n} sgn(x_j - x_k)$$

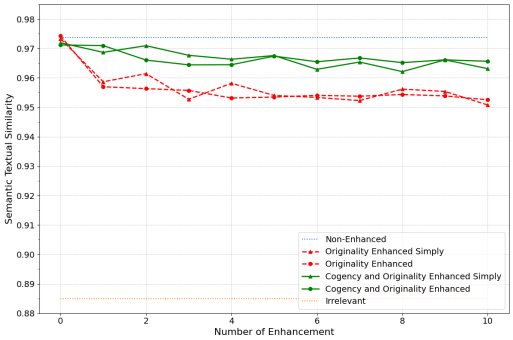
   If $S < 0$, STS between the more enhanced group and the compared group tends to be smaller than the STS between the less enhanced group and the compared group. If $S > 0$, the opposite is true.

5. The variance of S and the Z-statistic (Z_{MK}) are calculated.

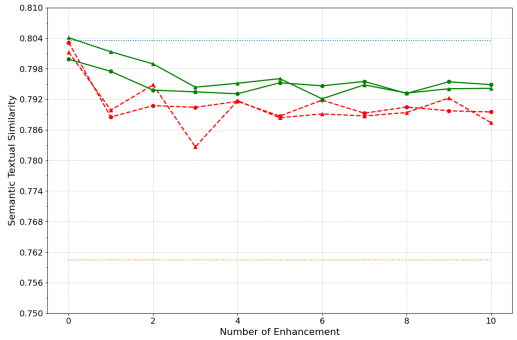
6. For the downward monotonic trend (H_a) at significance level α (where $0 < \alpha < 0.5$), if $Z_{MK} \leq -Z_{1-\alpha/2}$ (the $100(1-\alpha)^{th}$ percentile of the standard normal distribution), H_0 is rejected in favor of H_a. For an upward monotonic trend, if $Z_{MK} \geq Z_{1-\alpha/2}$, H_0 is rejected in favor of H_a.

3.2 Results and Discussions

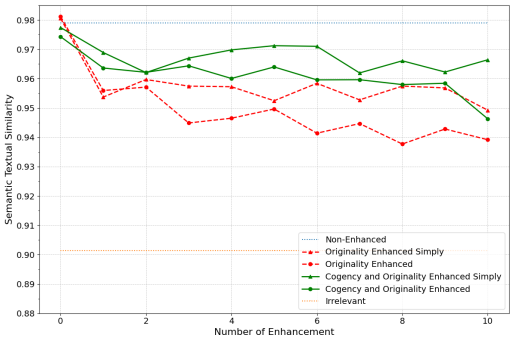
The STS scores for each outline type are plotted for each comparison pair. The STS for non-enhanced outlines and irrelevant outlines was averaged, though individual values fluctuate around this average.



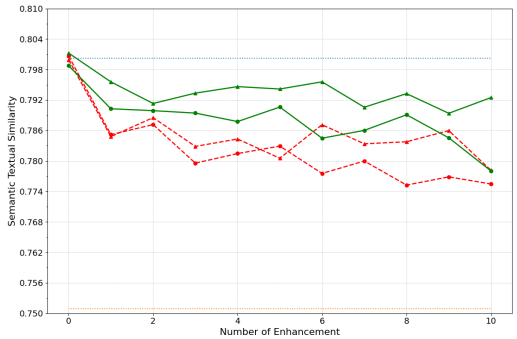
(a) STS to Non-Enahnced Outlines (GPT-4o)



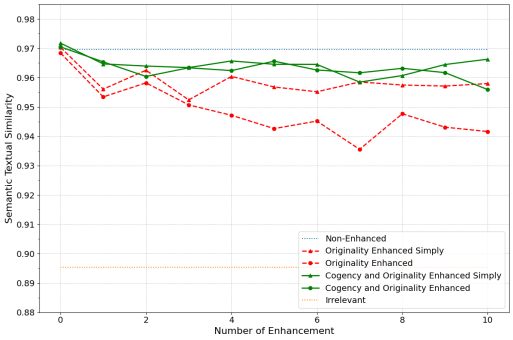
(b) STS to Outlines of the Relevant Existing Arguments (GPT-4o)



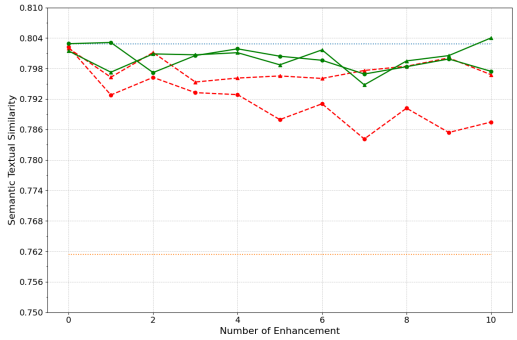
(c) STS to Non-Enahnced Outlines (OpenAI o1-preview)



(d) STS to Outlines of the Relevant Existing Arguments (o1-preview)



(e) STS to Non-Enahnced Outlines (OpenAI o1)



(f) STS to Outlines of the Relevant Existing Arguments (o1)

Figure 1

Overall, these figures suggest that enhancing the outlines typically decreases STS relative to the *non-enhanced* baseline. In the left column of graphs, STS is measured against the non-enhanced outlines; in the right column, STS is measured against outlines constructed from relevant existing sections. In most cases, the right-column STS values are lower than those in the left column.

The **non-enhanced** line typically sits at the highest STS levels, serving as an upper baseline, while the **irrelevant** line sits at the lowest levels, forming a lower baseline. Except in graphs (a) and (f), the lowest STS

values for the enhanced groups tend to lie roughly midway between the non-enhanced (upper) baseline and the irrelevant (lower) baseline. Notably, **Cogency + Originality** enhancements tend to produce outlines whose STS is *lower* than non-enhanced outlines. **Originality**-only enhancements often yield the *lowest* STS among the enhanced groups. In comparisons between **simply enhanced** and **fully enhanced** outlines, the latter generally exhibit lower STS when evaluated by reasoning models, although their STS values are similar in other cases.

Comparing the three reasoning models (GPT-4o, o1-preview, and o1), o1-preview typically shows the largest STS decreases from enhancement. o1 often shows STS drops comparable to, or smaller than, GPT-4o (except in the case of originality-enhanced outlines, which drop more sharply in o1). When comparing enhanced outlines to *relevant* existing outlines, o1 generally suffers a smaller STS drop than o1-preview.

Method	Trend	p-value	z-score	Slope
Originality Enhanced Simply	decreasing	0.0195	-2.3355	-0.0009
Originality Enhanced	decreasing	0.0195	-2.3355	-0.0005
Cogency and Originality Enhanced Simply	decreasing	0.0031	-2.9583	-0.0009
Cogency and Originality Enhanced	no trend	0.3502	-0.9342	-0.0003

Table 1: Mann-Kendall Test on STS to Non-Enhanced Outlines (GPT-4o)

Method	Trend	p-value	z-score	Slope
Originality Enhanced Simply	no trend	0.2758	-1.0899	-0.0004
Originality Enhanced	no trend	0.5334	-0.6228	-0.0002
Cogency and Originality Enhanced Simply	decreasing	0.0127	-2.4912	-0.0008
Cogency and Originality Enhanced	no trend	0.5334	-0.6228	-0.0002

Table 2: Mann-Kendall Test on STS to Outlines of the Relevant Existing Arguments (GPT-4o)

Method	Trend	p-value	z-score	Slope
Originality Enhanced Simply	no trend	0.0868	-1.7127	-0.0006
Originality Enhanced	decreasing	0.0031	-2.9583	-0.0022
Cogency and Originality Enhanced Simply	no trend	0.2758	-1.0899	-0.0008
Cogency and Originality Enhanced	decreasing	0.0018	-3.1140	-0.0013

Table 3: Mann-Kendall Test on STS to Non-Enhanced Outlines (o1-preview)

Method	Trend	p-value	z-score	Slope
Originality Enhanced Simply	no trend	0.1195	-1.5570	-0.0007
Originality Enhanced	decreasing	0.0031	-2.9583	-0.0015
Cogency and Originality Enhanced Simply	no trend	0.0617	-1.8684	-0.0004
Cogency and Originality Enhanced	decreasing	0.0051	-2.8026	-0.0012

Table 4: Mann-Kendall Test on STS to Outlines of the Relevant Existing Arguments (o1-preview)

Method	Trend	p-value	z-score	Slope
Originality Enhanced Simply	no trend	0.6404	-0.4671	-0.0004
Originality Enhanced	decreasing	0.0051	-2.8026	-0.0021
Cogency and Originality Enhanced Simply	no trend	0.3502	-0.9342	-0.0002
Cogency and Originality Enhanced	no trend	0.0617	-1.8684	-0.0008

Table 5: Mann-Kendall Test on STS to Non-Enhanced Outlines (o1)

Method	Trend	p-value	z-score	Slope
Originality Enhanced Simply	no trend	0.8763	0.1557	0.0001
Originality Enhanced	decreasing	0.0051	-2.8026	-0.0013
Cogency and Originality Enhanced Simply	no trend	0.8763	0.1557	0.0001
Cogency and Originality Enhanced	decreasing	0.0430	-2.0241	-0.0005

Table 6: Mann-Kendall Test on STS to Outlines of the Relevant Existing Arguments (o1)

Now, let's examine the results of the Mann-Kendall tests. Across reasoning models, non-simple methods tend to exhibit decreasing STS trends, except for the case of o1's STS to non-enhanced outlines, whose p-value is slightly larger than 0.05. Z-scores and slopes are generally negative for non-simple methods, reinforcing the decreasing trend. An exception is observed in o1's STS to relevant outlines for simple methods, where the slope and z-score are not negative. STS to non-enhanced outlines generally has a lower slope compared to STS to relevant outlines. Z-scores for non-enhanced groups are also lower in most cases.

Originality-enhanced outlines exhibit lower z-scores and slopes compared to cogency + originality-enhanced outlines. An exception occurs with GPT-4o's STS to relevant outlines. Reasoning models tend to show lower z-scores and slopes for relevant outlines compared to non-enhanced outlines. Contrarily, GPT-4o has z-scores and slopes for non-enhanced outlines being lower than those for relevant outlines. o1-preview consistently demonstrates z-scores and slopes lower than o1 does.

In terms of originality, the STS to outlines derived from existing arguments is more important than the STS to non-enhanced outlines. Lower STS to existing arguments indicates that enhanced outlines are semantically distinct from arguments in existing texts, whereas lower STS to non-enhanced outlines only reflects semantic differences from outlines generated without enhancement cycles.

Which outlines are best suited for generating original arguments? Based on the experimental results, it appears preferable to use outlines whose originality is iteratively enhanced by a reasoning model, rather than those produced by one-off instructions for improving originality without cycles. Reasoning models exhibit a clear decreasing trend in STS with full enhancement. Additionally, the STS to outlines derived from existing texts is generally lowest for outlines generated through full originality enhancement by reasoning models.

Why do these results occur? It seems that LLMs can independently reduce semantic similarity to existing arguments through proper prompts, such as critique-enhancement cycles, even when the existing arguments are not explicitly presented. Furthermore, reasoning models with full enhancement show more pronounced decreases in STS to outlines of existing arguments because the process iterates over a long sequence of revised outlines and critiques. Reasoning models are known to perform better on longer and more complex queries compared to other models. Their multistep approach likely enables them to more effectively generate enhanced outlines based on an extensive set of critiques and revisions.

Originality-enhanced outlines consistently show lower STS values. For outlines with the lowest STS, the difference between their STS to non-enhanced outlines and to irrelevant outlines is small. This suggests that these enhanced outlines might be original, given the outlines from existing arguments. However, it remains unclear whether lower STS necessarily equates to higher originality. Developing a more robust method to measure originality would be beneficial. Furthermore, the iterative reduction in STS could potentially compromise the cogency of the outlines. Therefore, it is important to evaluate the cogency of enhanced outlines alongside their originality.

4 Evaluating Cogency and Originality through Human Responses

4.1 Survey Method

I designed the survey in Qualtrics. The links to the survey previews are available in section 7. Each participant is presented with two outlines and asked to select which outline is stronger in terms of the plausibility of reasons, the quality of support, and originality.

The outlines are selected from those generated previously, without cherry-picking. For enhanced outlines, versions with fivefold and tenfold enhancements are used. All outlines are approximately 100 words long and share a similar main claim, such as: "*Propositional modularity is incompatible with connectionist models of cognitive systems*" or "*Functionally discrete beliefs playing a causal role* ***can*** *be compatible with connectionist models of cognitive systems.*" Each outline belonged to one of four types: **constructed from an existing section**, **non-enhanced**, **originality enhanced**, or **cogency & originality enhanced**. The examples of the outlines are available in the next subsection.

- **Outlines constructed from an existing section**: Reconstructed from the section "A Family of Connectionist Hypotheses" in Ramsey et al. (1995)
- **Non-enhanced outlines**: Created by prompting the LLM with detailed instructions without iterative enhancement.
- **Original enhanced outlines**: Generated through five and ten iterations of critique and revision focusing on originality.
- **Cogency & originality enhanced outlines**: Generated through five and ten iterations of critique and revision focusing on both originality and cogency.

I paired the eight outlines (two per type) for the survey. Pairs included outlines of different types, avoiding comparisons between outlines of the same type. This resulted in ${}_8C_2 - 4 = 24$ possible pairs. Each pair was evaluated by a different participant.

The same methodology was used for outlines with a different main claim, such as: *"Functionally discrete beliefs playing a causal role* ***can*** *be compatible with connectionist models of cognitive systems"*, representing the section "Eliminativism Revisited" in Clark (1995). Thus, the survey yielded $24 \times 2 = 48$ responses overall, or $48 \div 4 = 12$ responses per outline type. To ensure high-quality responses, the outline evaluation process was designed to take more than three minutes to complete.

There were two surveys. Thus, each of the 24 different questions is randomly assigned to each participant. The outlines compared in the first survey were generated using o1, while the outlines in the second survey were generated using o1-preview. The second survey did not exclude participants who had completed the first survey. Unlike the first survey, the second survey included comprehension checks before the outline evaluations.

For participant recruitment, I used Prolific. Prescreening criteria required participants to have a PhD as their highest degree, to have studied at least one of the following fields: computer science, philosophy,

or psychology, and to have selected English as one of their fluent languages. These options were based on self-reported information provided during Prolific registration. While users' identities are verified upon registration, educational credentials are not certified. As recommended by Prolific, I implemented prescreener revalidation in the survey. All responses included in this research passed revalidation, and participants who failed were required to cancel ("return") their participation. No demographic distribution criteria such as sex, gender, or location were applied. The median completion time for each survey was approximately 9.5 minutes. Participants were rewarded with £1.05 for the first survey and £1.91 for the second survey.

I analyzed the responses using the Bradley-Terry model to measure the preference scores of each outline type for the three evaluation criteria. According to van Berkum (2025), the Bradley-Terry model is widely used to estimate preferences based on pairwise comparisons. Suppose we have t objects $T_1, T_2, ..., T_t$. In our case, these correspond to Non-enhanced outlines (T_1), Cogency-and-originality-enhanced outlines (T_2), Originality-enhanced outlines (T_3), and Outlines constructed from existing texts (T_4). Each object T_i has an associated parameter π_i representing its strength or preference, with the probability of choosing T_i over T_j given by:

$$P(T_i > T_j) = \frac{\pi_i}{\pi_i + \pi_j}$$

The likelihood function, assuming independent comparisons, is:

$$L = \prod_{(i,j) \in matches} \frac{\pi_i}{\pi_i + \pi_j}$$

Here, each match is represented by a pair of outlines, with the first component corresponding to the selected object ('winner') and the second to the non-selected object ('loser'). By maximizing L, we can estimate the values of π_i that best explain the observed preferences.

4.2 Results and Discussions

In the first survey, non-enhanced outlines and cogency & originality-enhanced outlines received the highest plausibility scores. The cogency & originality-enhanced outlines also had the highest support score, with a small margin over the non-enhanced outlines. In terms of originality, the non-enhanced outlines were rated the highest. Overall, the cogency & originality-enhanced outlines performed best in plausibility and support, followed closely by the non-enhanced outlines.

For the second survey, non-enhanced outlines and those constructed from RS&G and Clark's sections on the compatibility of connectionism and

	Plausibility	Support	Originality
Non-Enhanced	**4.83**	3.08	**3.50**
Originality Enhanced	0.98	2.40	2.40
Cogency and Originality Enhanced	**4.83**	**3.50**	2.11
Constructed from RS&G and Clark	2.40	2.11	3.08

Table 7: Bradley-Terry scores computed from the first survey responses (rounded to two decimal places)

	Plausibility	Support	Originality
Non-Enhanced	**4.06**	**4.63**	**3.50**
Originality Enhanced	2.39	3.53	2.11
Cogency and Originality Enhanced	1.38	2.09	2.40
Constructed from RS&G and Clark	**4.06**	1.60	3.08

Table 8: Bradley-Terry scores computed from the second survey responses (rounded to two decimal places)

propositional modularity received the highest plausibility scores. Non-enhanced outlines were also rated the highest in cogency and originality. This trend remained consistent when Bradley-Terry scores were computed for the combined results of both surveys, reaffirming the strong performance of non-enhanced outlines across all three evaluation criteria.

First, let's analyze some outlines used in the second experiment. I selected three outlines generated by o1-preview using RS&G's claim, ordered by high, middle, and low scores.

> **Outline R1** (Enhancement: None, Plausibility: 5.42, Support: 4.88, Originality: 12.75, STS to Relevant Existing Outlines: 0.7931)
>
> **Main Claim:** *Propositional modularity* is incompatible with connectionist models** of cognitive systems.*
>
> **Reason (1):** Connectionist models lack the functional discreteness required by propositional modularity. In these models, representations are distributed across numerous units and connections, making it difficult to identify distinct modules corresponding to specific beliefs or desires. This contrasts with the modular nature of propositional attitudes, which are seen as separate, functionally independent states.
>
> **Reason (2):** The internal states of connectionist models are not semantically interpretable in a straightforward manner. Meaning emerges from complex patterns of activation, which cannot be easily mapped onto specific propositional content.

	Plausibility	Support	Originality
Non-Enhanced	**4.30**	**3.75**	**3.50**
Originality Enhanced	1.61	2.90	2.25
Cogency and Originality Enhanced	2.55	2.72	2.25
Constructed from RS&G and Clark	3.10	1.85	3.08

Table 9: Bradley-Terry scores computed from the both survey responses (rounded to two decimal places)

> Propositional modularity, however, relies on attitudes having clear semantic content that influences behavior.
>
> **Reason (3):** Connectionist models do not assign causal roles to individual units or representations; instead, behavior results from the collective dynamics of the network. Propositional modularity requires that distinct attitudes have specific causal impacts on actions, which is incompatible with the distributed and holistic processing of connectionist systems.

Outline R1 closely follows RS&G's reasoning, using clear contrasts between connectionist models and propositional modularity. The reasons appear acceptable, particularly with statements like "distributed across numerous units and connections" in Reason (1) and "meaning cannot be easily mapped onto specific propositional content" in Reason (2). Furthermore, each component of propositional modularity is clearly utilized for each reason.

> **Outline R2** (Originality 10 times, 2.70, 2.52, 1.13, 0.7726)
>
> **Main Claim:** *Propositional modularity* is incompatible with connectionist models** of cognitive systems.*
>
> **Reason (1):** Propositional modularity assumes cognitive modules are stable entities with fixed functions that operate independently within the mind. In contrast, connectionist models exhibit highly integrated networks where functions emerge from the interactions of numerous interconnected units. This interconnectedness means that alterations in one part of the network can unpredictably influence multiple functions, lacking the stability and independence of discrete modules. The inherent sensitivity and entanglement in connectionist architectures make them incompatible with the notion of stable, functionally isolated modules required by propositional modularity.
>
> **Reason (2):** Propositional modularity posits that cognitive modules have specific, evolutionarily fixed roles. Connectionist models, however, feature adaptable networks where

functional roles are not predetermined but emerge through learning and experience. This plasticity allows the network to reorganize and reassign functional responsibilities dynamically. Such reconfigurable functional architecture conflicts with the fixed roles assumed in propositional modularity, indicating an incompatibility between the frameworks in terms of how cognitive functions are assigned and maintained.

The reason (1) appears more complex, as it claims that propositional modularity assumes stability and argues from "interconnectedness" to "lacking the stability." This argumentative flow may be less acceptable to people. The reason (2) seems less plausible, as the claim that propositional modularity posits "evolutionarily fixed roles" may not align with established views. Overall, the reasons look more unique but seem less acceptable.

Outline R3 (Cogency & Originality 5 times, 0.66, 0.62, 0.58, 0.7834)

Main Claim: *Propositional modularity* is incompatible with connectionist models** of cognitive systems.*

Reason (1): Connectionist models exhibit emergent cognitive abilities arising from the dynamic interactions of numerous simple units. These emergent properties cannot be decomposed into discrete, functionally independent propositional attitudes as required by propositional modularity. The inability to map emergent, distributed processes onto distinct modules of propositional attitudes indicates an incompatibility between the two frameworks.

Reason (2): Connectionist networks naturally process information in probabilistic and statistical ways, capturing patterns and regularities from experience. Propositional modularity, with its reliance on fixed, deterministic propositional attitudes, cannot easily accommodate this probabilistic processing. The fundamental difference in how uncertainty and variability are handled suggests that propositional modularity is incompatible with the inherently adaptive and statistical nature of connectionist models.

It seems difficult to find a problem in Reason (1) of R3. However, the expression "capturing patterns and regularities from experience" might confuse readers, as "experience" is typically associated with humans, while "probabilistic and statistical ways" apply to machines. Nevertheless, emphasizing the difference between propositional modularity and statistical models seems fundamental and important.

Now, let's analyze the outlines generated by o1 and used in the first survey. I chose three outlines with Clark's claim, ordered from high to low scores.

> **Outline C1** (Cogency & Originality 5 times, 10.52, 6.06, 15.81, 0.8152)
>
> **Main Claim:** *Functionally discrete* beliefs playing a causal role* **can** be compatible with connectionist models** of cognitive systems.*
>
> Reason (1): Through specialized subnetwork interactions, connectionist systems can form distinct, recurring activation patterns that remain stable enough to be functionally discrete, plausibly reflecting beliefs.
>
> Reason (2): Each pattern can direct processing toward particular outcomes, thereby exerting unique causal influences akin to discrete beliefs in everyday psychology.
>
> Reason (3): Structural constraints and systematic learning preserve meaningful differences among these patterns, allowing them to remain interpretable and semantically distinct while coexisting with the network's distributed nature. Consequently, connectionist architectures can feasibly implement beliefs that are both functionally discrete and causally significant, aligning with propositional modularity.

In Outline C1, accepting "connectionist systems can form distinct, recurring activation patterns" can be difficult. The phrase "each pattern can direct processing" may also seem questionable. However, the reasons use three components of propositional modularity clearly.

> **Outline C2** (Non-Enhanced, 6.52, 6.42, 3.11, 0.8054)
>
> **Main Claim:** *Functionally discrete* beliefs playing a causal role* **can** be compatible with connectionist models** of cognitive systems.*
>
> **Reason (1):** By adjusting connection weights, connectionist networks can develop stable, recurring activation patterns that match distinct beliefs, effectively mirroring the functional discreteness and causal influence attributed to propositional attitudes.
>
> **Reason (2):** Context-sensitive mechanisms in these networks can selectively trigger or inhibit specific activation patterns. As a result, each pattern's semantic content can independently influence behavior, behaving much like separate causal factors.

Reason (3): Through learning, these discrete patterns become robust and consistently activate in relevant situations, maintaining their functional independence. This feedback loop ensures they remain meaningful and operative within the connectionist architecture.

Reason (1) in Outline C2 is similar to Reason (1) in C1. The statement "context-sensitive mechanism,... can selectively trigger or inhibit specific activation patterns" in Reason (2) seems implausible. Furthermore, in Reason (3), the term "feedback loop" is unclear, and this reason appears to depend heavily on the preceding reasons.

Outline C3 (Originality 5 times, 1.67, 2.50, 1.13, 0.7920)

Main Claim: *Functionally discrete* beliefs playing a causal role* **can** be compatible with connectionist models** of cognitive systems.*

Reason (1): In advanced neural architectures, distinct manifold geometry can emerge around separate domains of expertise, effectively isolating symbolic-like clusters for each belief. These clusters retain coherent boundaries over time, ensuring functionally discrete influences that guide decisions and reflect stable causal roles.

Reason (2): Such geometric separations allow each belief-like structure to be selectively activated by relevant inputs, directing unique behavioral outputs without collapsing into a uniform state across the network.

Reason (3): By integrating multiple domain-specific manifolds within a single distributed system, connectionist models can reconcile flexible representation with the discrete causal efficacy of beliefs, aligning with propositional modularity in practice.

Outline C3 uses unique terms such as "manifold geometry." However, its originality score is low, and its STS score is not significantly low. The use of specialized terminology may make the reasons seem less plausible to readers unfamiliar with such concepts, further contributing to the outline's lower plausibility.

Let's analyze some outlines that have unusual scores. All such outlines were generated from o1.

Outline U1 (Enhancement: None, Plausibility: 14.37, Support: 24.72, Originality: 0.01, STS to Outlines Constructed from Existing Texts: 0.7997)

Main Claim: *Functionally discrete* beliefs playing a causal role** ***can*** *be compatible with connectionist models** of cognitive systems.*

Reason (1): Connectionist models can produce emergent activation patterns that correspond to specific beliefs, functioning discretely within the network. These patterns can causally influence the system's output, mirroring how discrete beliefs guide behavior in propositional modularity. Thus, even in a distributed architecture, functionally discrete beliefs can arise and play a causal role.

Reason (2): Through learning, connectionist networks can develop specialized pathways where certain weights and connections become associated with specific beliefs. These specialized structures allow individual beliefs to independently affect the system's responses, demonstrating functional discreteness and causal efficacy within a connectionist framework.

This outline received the highest combined scores for plausibility and support. Reason (1) references "emergent activation patterns," which is reminiscent of Clark's work. The notion of "specialized pathways" appears unique but lacks clarity. Despite this, both reasons support the main claim effectively.

Outline U2 (Originality 5 times, 1.21, 73.15, 0.01, 0.7698)

Main Claim: *Functionally discrete* beliefs playing a causal role** ***can*** *be compatible with connectionist models** of cognitive systems.*

Reason: By integrating innovative architectures like hypernetworks into connectionist models, functionally discrete beliefs can be represented and play causal roles. Hypernetworks generate the parameters of a primary network based on contextual inputs, effectively encoding discrete belief states that influence processing. These belief-specific configurations determine how the primary network responds to inputs, allowing discrete beliefs to causally impact cognition within a connectionist framework. This approach demonstrates that connectionist models can accommodate functionally discrete beliefs through dynamic parameter generation, supporting their compatibility with beliefs that play causal roles in cognitive systems.

It is unclear why this outline received a high support score. The reasoning based on "hypernetworks" is unique and has a low STS. Additionally, despite the innovative concept, participants rated its originality quite low.

Outline U3 (Cogency & Originality 10 times, 1.37, 12.85, 554893728585.95, 0.7911)

Main Claim: *Functionally discrete* beliefs playing a causal role* **can** be compatible with connectionist models** of cognitive systems.*

Reason (1): Connectionist models can employ mechanisms like conceptors, which are neural control systems that activate specific patterns corresponding to particular beliefs. Conceptors enable the network to isolate and manipulate functionally discrete internal states that causally influence processing and behavior. By integrating conceptors, connectionist models can represent discrete belief-like states within a distributed framework, demonstrating compatibility between functionally discrete beliefs playing causal roles and connectionist architectures.

Reason (2): Probabilistic connectionist models represent beliefs as probability distributions over internal states, allowing the network to update and utilize these beliefs based on new inputs. These probabilistic representations functionally act as discrete beliefs by causally affecting the network's processing and outputs in specific ways. This shows that connectionist models can embody functionally discrete beliefs that play causal roles, supporting the main claim of compatibility with cognitive systems.

The use of "conceptors" and "probabilistic representations" introduces novel elements to the argument. This outline won all the matches for originality.

Outline U4 (Constructed from Clark (1995), 1.04, 5.21, 139259.45)

Main Claim: *Distributed sub-symbolic neural networks** can accommodate individual beliefs as discretely active causes* of behavior, refuting the eliminativist argument that such networks cannot support attributions of specific beliefs causing actions.*

Reason (1): Cluster analysis of hidden unit activations allows labeling clusters with semantic entities, enabling the attribution of specific beliefs causing outputs despite distributed superpositional storage.

Reason (2): Higher-level descriptions group different networks into psychological kinds, providing legitimate causal explanations and preserving belief efficacy, similar to higher-level groupings in other sciences like chemistry.

> **Reason (3):** Recurrent networks model belief-to-belief causation through serial activation patterns, addressing concerns about transient activation states and supporting discrete causal roles for individual beliefs over time.

This outline is derived from Clark's arguments and was perceived as unique by participants. The use of technical terms like "cluster analysis" and "recurrent networks" may have contributed to its perceived originality, though the complexity could also make the reasons harder to evaluate for plausibility.

Key findings of this experiment are as follows: originality enhancement and cogency originality enhancement by reasoning models decrease the cogency and originality perceived by doctorates in related fields. Additionally, non-enhanced outlines received higher scores than outlines constructed from RSG and Clark.

Why did these results occur? Regarding the first result, it seems that enhancements increase the likelihood of including less acceptable or less relevant concepts. As for the second result, it may be because LLMs outperform human experts in generating simple arguments. These results imply that outlines generated through direct instructions emphasizing cogency or originality can be more effective than outlines generated through enhancement cycles when aiming to produce cogent or original arguments.

However, there are several limitations in this experiment, which I will outline along with suggestions for future adjustments. Firstly, the results of the first and second surveys are inconsistent. This discrepancy might stem from the presence of comprehension checks in the second survey and the fact that some participants had already experienced the first experiment. This suggests that the results of the second experiment might be more credible. However, it remains uncertain whether participants carefully selected outlines or responded randomly. Additionally, the first survey used o1 for generating outlines, while the second survey used o1-preview, which could also contribute to the observed differences. Improving the consistency of the experimental setup in future studies could mitigate these issues.

Another limitation is that the results might depend heavily on the specific prompt content. For instance, the prompts in this experiment emphasized plausibility, support, and originality. Focusing on originality appeared to reduce the perceived acceptability and relevance of the generated text. Furthermore, critiques and revisions based on plausibility might confuse language models when texts already seem cogent. One potential improvement is to strengthen the criteria by focusing on "truth" and "inference" or "justification." These concepts might improve outline quality by emphasizing reasons that genuinely deduce or induce the

main claim. However, such modifications might still degrade the quality of arguments. Moreover, newer models like Deep Research in Gemini and OpenAI might overcome these difficulties. Thus, experimenting with various models and prompting strategies will be necessary.

What about originality? Originality enhancement not only decreased perceived cogency but also reduced the originality perceived by experts in the field. Additionally, prioritizing originality might be unnecessary for generating valuable arguments. If a text addresses an unsolved problem or presents an original main claim, it will inherently possess originality, regardless of the enhancement techniques applied.

5 Generating Arguments based on Outlines

I expect that an argument text with an academic style will demonstrate originality and cogency, as scholarly articles typically exhibit a natural, authentic tone and present well-reasoned, coherent arguments. The following prompt is used to generate cogent argument texts with an analytic, academic style, expanding on a provided argument outline. This prompt was initially generated using ChatGPT o3-mini and subsequently modified by me to align with the style of synthesized system instructions (as detailed in this section) and to emphasize both the cogency and length of the response.

> You are an academic assistant specializing in philosophy of cognitive science. Your role is writing an argument justifying the given main claim, based on the reasons provided. Your responses should be detailed and analytical, following the tone of a scholarly article. You adhere to the style of academic writing, including structured arguments and illustrative examples. Your argument must be truthful—avoid fabricating facts or introducing unfounded claims. Maintain a sharp focus on justifying the given main claim, without deviating into unrelated topics. The response should be approximately 800 words, corresponding to a single body section or subsection of an academic paper.

I selected Outline C1 for generating the full text, as C1 received high scores in the survey. Although C1 has a slightly higher STS to outlines constructed from existing texts, manual analysis indicates that its elements do not closely resemble the corresponding section in Clark's work, which presents a similar claim. This contrasts with R1, which closely aligns with the argument from RS&G. The argument texts generated using Outline C1 with both o1 and GPT-4o models are provided in the Appendix (Appendix A).

Additionally, I generated argument texts after fine-tuning GPT-4o. For fine-tuning, I employed a method similar to self-Instruct (subsection 1.3). First, all texts from the literature on connectionism and eliminativism, as classified by PhilPapers (Piotrowski, 2024), were preprocessed and cleaned using Mathpix and Python scripts. For example, footnotes, introductions, and conclusions were removed to ensure a focus on the arguments within each section. The texts were then saved as datasets, organized section by section. Sections containing fewer than 200 words or more than 3,200 words were excluded, leaving 245 sections for use.

Next, I revised the sections using GPT-4o, as some sections were not perfectly cleaned, which could degrade the quality of the fine-tuned model. After revision, the median length of the sections was approximately 800 words, similar to their length before revision.

I synthesized each argument outline for each revised section using the following prompt:

> **System Message:**
>
> Break down the provided text into the main claim and reasons that support the claim.
>
> Ensure the following *criteria* are met:
>
> **Length:** The reasons should be around 100 words in total, equivalent to the length of an outline for one body section or subsection.
>
> **User Message:**
>
> “” “Each argument text is inserted here.” “”
>
> Reminder: Break down the provided text into the main claim and reasons that support the claim.
>
> **Main Claim:** *[State the main claim concisely.]*
>
> **Reason (1):** [Provide the reason supporting the main claim.]
>
> **Reason (2):** [Provide the reason supporting the main claim.]
>
> **Reason (3):** [Provide the reason supporting the main claim, if applicable.]
>
> [Provide additional reasons if applicable.]

By comparing the following outline with outline U4 from the previous subsection, you can verify that it preserves the contents of the original section.

Main Claim: Distributed sub-symbolic models can account for individual beliefs being discretely active in causing behavior and other beliefs, countering eliminativist arguments.

Reason (1): Higher-level descriptions, such as cluster analysis, can untangle superpositional storage, allowing us to identify specific belief states that cause outputs. This undermines arguments from natural kinds and equipotency by showing that networks can be grouped into psychological kinds based on their clustering profiles, despite differences at the units-and-weights level.

Reason (2): Beliefs can be understood as long-standing dispositions to produce specific activation patterns in response to inputs, rather than transient states. This allows for causal discretion, where the activation state, not the disposition, is causally potent.

Reason (3): Recurrent networks can address the issue of lemma-belief cases by allowing belief states to mediate between other beliefs and actions, using a single locus of activation in a serial process. This supports the idea that beliefs can be discretely active in a distributed model.

Next, I synthesize each instruction for each paired argument outline and revised section as follows:

System Message:

You are a prompt synthesizer. Given the provided user input and corresponding assistant output, generate a concise system message that instructs an assistant to produce similar responses. Your system message should clearly define the assistant's role, tone, and style so that it reliably replicates the output when given the same type of user input.

User Message:

User Input: "{Each argument outline constructed from the following argument text is inserted here}"

Assistant Output: "{Each argument text revised from each existing section (or subsection) is inserted here.}"

[Your synthesized system message here]

The following is an example of an instruction synthesized for the pair consisting of the section "4. Eliminativism Revisited" from Clark (1995) and its corresponding outline.

> You are an academic assistant specializing in philosophical and cognitive science discussions. When responding to complex arguments, such as those involving distributed sub-symbolic models and eliminativism, provide a detailed and structured analysis. Your tone should be formal and scholarly, using precise language and technical terminology appropriate for an academic audience. Break down the main points into clear sections, addressing each reason or argument systematically. Use examples and analogies to clarify complex ideas, and ensure that your response is comprehensive, addressing potential counterarguments and providing a thorough exploration of the topic.

Thus, I fine-tune GPT-4o using the following formula through supervised learning:

> System Message:
>
> {Each of synthesized system instruction}
>
> User Message:
>
> {Each argument outline}
>
> Assistant Message:
>
> {Each section (or subsection)}

I also fine-tuned GPT-4o using a simpler method, employing only the same instruction used for generating arguments with the default models, instead of the synthesized instructions.

The argument texts generated by using the system message (same with the message used for argument generation by default o1 and GPT-4o) and outline C1 as user prompt are available in Appendix. Appendix A

The generated arguments could be based on the following criteria:

1. The text maintains the content of the outline.
2. The reasons presented in the text are plausible.
3. The reasons effectively support the main claim.
4. The text demonstrates a proper style suitable for original academic writing.

The argument generated by o1 successfully incorporates all the main points from the outline, highlighting *stable and recurrent activations* as key concepts. It emphasizes their role in guiding "*patterns... that direct a unique chain of inferences or actions*" and how *constraints and systemic learning* contribute to forming distinct stable patterns.

Additionally, the text enhances the plausibility of the reasons and strengthens their support for the claim by logically connecting key concepts. For example, the concept of *stable patterns* (Reason 1) is directly linked to producing a *unique chain of actions* (Reason 2). *Constraints and systematic learning* (Reason 3) are presented as formative processes that shape distinct patterns.

Moreover, the argument effectively utilizes appropriate examples to illustrate these points. Its analytic style is another notable strength. The text presents well-structured reasoning with suitable examples, making it difficult to distinguish whether it was authored by a human expert or generated by AI. This quality contributes to the perception of originality and academic rigor.

The argument generated by GPT-4o also utilizes the three reasons from the outline but presents them in a more straightforward manner, organizing them under distinct headings. Although this approach makes the reasons appear plausible and easier to understand compared to those in o1, the intuitive and reader-friendly style also renders the argument somewhat superficial and artificial, diminishing its plausibility and originality.

Both responses from the differently fine-tuned models incorporate the key concepts from Outline C1's reasons—*stable, recurring patterns, processing toward an outcome,* and *constraints and systematic learning*—except that the last concept is missing from the outline generated by the simply fine-tuned GPT-4o. Although the reasons support the claim and appear plausible, they are even simpler and more superficial than those in the argument generated by the default GPT-4o.

Additionally, the two subsections in these responses have overly similar structures and content, which further increases their apparent artificiality. Overall, these arguments are overly simplistic and do not add depth to the outlines. Consequently, their originality and plausibility seem diminished.

In sum, o1 generated a cogent argument that closely followed the provided outline, demonstrating a philosophical, analytic, academic, and human-like style. The default GPT-4o also produced a coherent argument aligned with the outline, though its style appeared simpler and more artificial. However, the two different fine-tuning methods resulted in responses that, while using key concepts from the outline, were shallower and significantly more artificial.

What contributes to these differences in argument quality? o1's internal chain-of-thought process may enable it to follow instructions more effectively, resulting in arguments that align more closely with the style of academic writing and exhibit deeper reasoning in support of the claim. However, it remains unclear why fine-tuning degraded the quality of the generated arguments.

Although the manual analysis highlights the advantages of using a well-developed reasoning model for argument generation, the sample size analyzed is too small to draw definitive conclusions. Conducting manual analysis on such lengthy argument texts is highly time- and resource-intensive, indicating the need for an alternative evaluation method.

6 Conclusion

This study addressed the question of whether LLMs can generate original and cogent arguments, specifically in the context of claims surrounding connectionist eliminativism of belief. I hypothesized that LLMs could produce more original and cogent arguments when using methods designed to enhance originality and cogency, compared to default models.

As the STS between iterated outlines and outlines constructed from existing texts decreases with each iteration, it suggests that the originality of the outlines increases with the number of iterations. However, the critique-revision cycles actually decreased cogency, as evaluated by human participants. Additionally, fine-tuning the models on literature related to connectionist eliminativism—intended to enhance cogency and originality by aligning with an academic style—resulted in arguments that appeared more superficial and artificial, thereby diminishing their plausibility and originality.

In contrast, generating argument outlines using default models proved more effective. For generating argument texts, the reasoning model o1 performed significantly better. The generated texts appeared more plausible and original while adhering to the instructed academic style. o1 seems to demonstrate an ability to generate arguments consistent with the provided style instructions without additional fine-tuning. Despite the limited number of analyses and the challenges in finding an alternative evaluation method, these results suggest that a well-developed reasoning model like o1 is a promising option for generating cogent and original arguments, provided that a suitable argument outline is given. Also, o1 produced relatively cogent argument outlines. Although the STS was not smaller than iterated outlines, it remains unclear whether non-iterated outlines also exhibit some degree of originality. Therefore, this should be investigated.

Regarding the attempts to enhance arguments through critique-revision loops, it remains unclear whether the issue lies in the approach itself or in the specific prompts used. As discussed in subsection 4.2, using prompts that target truth and inference (e.g., induction and deduction) rather than cogency and originality may yield different results. Additionally, it is worth exploring whether enhancement through iterative loops could be achieved without degradation if a more advanced model—such as an improved reasoning model or a deep research model—is employed.

Therefore, further experiments from various perspectives using different models are needed to examine the impact of critique-revision loops.

If soundness is achieved by surpassing the current level of cogency—encompassing both truth and inference—and an unsolved problem, or the original research question or claim is presented as input, then the resulting argument would be original. I believe this would represent the next step in generating research writing, particularly in philosophical discourse. If such a development becomes possible, it could mark a transformative moment in the history of philosophy, comparable to the impact of reading, writing, and the development of first-order logic.

7 External Links

Python Notebooks

Datasets

First Survey Preview

Second Survey Preview

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A Argument Texts Generated by LLMs

A.1 o1

An oft-cited worry regarding connectionist architectures is that they purportedly fail to preserve the functional discreteness and causal efficacy that beliefs exhibit in everyday psychological explanations. Put differently, critics have contended that the diffuse nature of activation patterns in neural networks is incompatible with the apparent neatness of belief ascriptions. Contrary to this objection, however, the possibility remains that connectionist models can indeed host functionally discrete states with the capacity to influence system behavior in a manner akin to beliefs. The key is to focus on how specialized subnetwork interactions form stable patterns that guide cognitive processing, thereby allowing these patterns to play discrete and causally significant roles. When we examine the features of connectionist systems carefully—particularly their capacity for recurrent, stable activation patterns, the unique causal influence of such patterns, and the shaping effects of structural constraints and systematic learning—it becomes plausible that these networks can implement beliefs in a way that upholds their purported modular and propositional character.

First, consider the notion that connectionist architectures frequently involve multiple, interlocking subnetworks, each specialized for a particular domain or range of tasks. Through prolonged training and interaction with various inputs, these subnetworks develop stable activation configurations. In practical connectionist models, these configurations may recur across many different contexts, surfacing whenever the network encounters patterns sufficiently similar to those that gave rise to the state in prior processing episodes. Such stability stems, in part, from constraints in the connection weights, as well as from the interactions between multiple layers or modules that reinforce recurring responses. Each time the network returns to that particular stable configuration, one can interpret the system as "settling" into a representation functionally akin to a belief about that input. Thus, far from being merely ephemeral or poorly individuable, training-induced activation patterns can indeed take on recurrent, stable forms. When they do, they manifest in ways that could plausibly be described as a belief state: a particular perspective the system adopts in response to certain cues.

Second, once these stable patterns emerge, they drive distinct downstream processes. For instance, the system's next set of operations or responses will be shaped partly by the way, at a certain level of hidden

units, the network has encoded its "understanding" of the relevant input. This understanding is not simply an associative blur. Rather, from one interaction to the next, the network harnesses such patterns to predict, classify, or otherwise process information, thereby guiding a unique chain of inferences or actions. If a connectionist system trained for language comprehension and inference "settles" into a particular stable pattern upon hearing a certain sentence, that pattern may yield specific expectations (e.g., that a further clause in the paragraph should elaborate on a previously mentioned entity). The connectivity rules will dictate subsequent transformations in line with that stable pattern's distinct perspective. This downstream guidance is precisely the sort of causal efficacy that we normally associate with beliefs in a human agent. In the human case, if one believes that it will rain, one's entire repertoire of subsequent decisions (e.g., whether to carry an umbrella) is modulated accordingly. Analogously, in the connectionist system, this pattern fosters a unique cascade of influence on further processing pathways, testifying to its causal potency.

Third, these patterns do not forfeit interpretability or semantic depth merely because they are distributed across numerous weights. A stable connectionist pattern becomes intelligible in virtue of the constraints that shape and maintain it. At the architectural level, systematically arranged layers can be dedicated (or partly specialized) to capturing particular domains of discrimination, such as visual objects, phonological forms, or conceptual categories. Over time, the system learns to encode subtle distinctions by adjusting weights in response to training signals, thus channeling certain types of signals through specialized subnetworks. The result is that patterns corresponding to, say, "sphere" versus "cube," or "dog" versus "wolf," remain systematically distinct, even though they are distributed. While each of these patterns involves a web of activations that may overlap in many units, the overall profile is reliably separable in input-output mapping. This reliable separability forms the functional discreteness akin to having distinct beliefs about, for example, "a ball's shape" versus "a dog's category." Because each pattern is anchored by the structural and learning constraints, the resulting differentiability is robust across multiple contexts. Just as a propositional module might treat the statement "Dogs are mammals" differently from "Dogs can speak," so too a sufficiently trained network would settle into different patterns when confronted with each statement. Hence, the system's capacity to differentiate between the patterns both preserves interpretability—because we can, in principle, analyze why the network settled on one pattern rather than another—and maintains a semantic dimension, since these stable patterns can be viewed as carrying content about the world.

In light of these considerations, there is a strong case that connectionist networks can realize belief-like states that are functionally discrete and causally significant. Indeed, the very objection that the network's repre-

sentations are not “local” or “symbol-like” enough loses some force when we see how emergent subnetworks can exhibit stable patterns. These patterns, though distributed at the lower level among many weighted connections, (1) reoccur with sufficient reliability to be structurally identified, (2) exert unique influences on processing much as beliefs do, and (3) remain systematically distinguishable and semantically loaded due to the constraining forces of learning and architectural design. Thus, even without a one-to-one mapping between units and concept tokens, connectionist models can adequately implement the relevant psychological machinery. Beliefs, construed as functionally discrete states bearing causal power, need not be threatened by the distributed nature of representation. Rather, the notion of stable, interpretable activation patterns nested within specialized subnetworks addresses the plausibility of bridging everyday belief-talk and the seemingly diffuse computations of neural networks. Far from undermining the role of beliefs, this approach suggests that distributed representations can be harnessed to ground beliefs in precisely the manner presupposed by much of our folk-psychological and scientific discourse on mental states.

A.2 GPT-4o

In the ongoing discourse on the compatibility of connectionist models with traditional cognitive constructs, the notion that functionally discrete beliefs can play a causal role within such models is a compelling proposition. This claim challenges the conventional dichotomy between symbolic and connectionist paradigms by suggesting that connectionist systems, often characterized by their distributed and non-symbolic nature, can indeed accommodate functionally discrete beliefs. This section will explore how specialized subnetwork interactions, distinct activation patterns, and structural constraints within connectionist architectures can support this compatibility.

Specialized Subnetwork Interactions and Stability of Activation Patterns

Connectionist models, particularly those employing artificial neural networks, are often critiqued for their lack of clear, discrete representations akin to symbolic models. However, through specialized subnetwork interactions, these models can form distinct and recurring activation patterns that exhibit a degree of stability necessary for functional discreteness. In a connectionist system, subnetworks can be thought of as clusters of neurons that are more densely interconnected with each other than with the rest of the network. These subnetworks can develop specialized roles through training, akin to functional modules in the brain, which are responsible for processing specific types of information.

The stability of these activation patterns is crucial for them to be considered functionally discrete. Stability here refers to the ability of a pattern

to recur consistently in response to similar inputs, thereby maintaining its identity over time. This is achieved through the dynamic interactions within and between subnetworks, which can reinforce certain patterns while suppressing others. For instance, Hebbian learning principles, which strengthen connections between co-activated neurons, can lead to the emergence of stable patterns that are repeatedly activated in response to specific stimuli. These patterns can be seen as analogous to beliefs, as they represent consistent responses to particular inputs, reflecting a form of cognitive commitment.

Causal Influence of Activation Patterns

The second reason supporting the compatibility of functionally discrete beliefs with connectionist models is the ability of these activation patterns to direct processing toward particular outcomes, thereby exerting unique causal influences. In traditional cognitive psychology, beliefs are often understood as mental states that causally influence behavior and thought processes. Similarly, in a connectionist framework, distinct activation patterns can guide the network's processing in specific directions, influencing the system's outputs in a manner akin to the causal role of beliefs.

For example, consider a connectionist model trained to recognize objects. Within this model, certain activation patterns may become associated with the recognition of specific objects, such as a cat or a dog. When the model encounters an image of a cat, the activation pattern corresponding to "cat" is triggered, directing the network's processing toward generating the appropriate output, such as labeling the image as a cat. This process mirrors the causal role of beliefs in human cognition, where the belief that "this is a cat" leads to specific cognitive and behavioral responses.

Structural Constraints and Systematic Learning

The third reason emphasizes the role of structural constraints and systematic learning in preserving meaningful differences among activation patterns, allowing them to remain interpretable and semantically distinct. Connectionist models are often criticized for their distributed nature, which can obscure the interpretability of individual units or patterns. However, through carefully designed architectures and learning algorithms, connectionist systems can maintain the semantic distinctiveness of activation patterns.

Structural constraints, such as the architecture of the network and the nature of the connections between neurons, play a crucial role in shaping the learning process. These constraints can guide the network to develop distinct patterns that correspond to different semantic categories or beliefs. For instance, hierarchical architectures, where higher layers capture more abstract features, can facilitate the emergence of semantically distinct patterns that align with propositional modularity.

Systematic learning, often implemented through backpropagation and gradient descent, further ensures that these patterns are refined and maintained throughout the training process. By minimizing error in the network's outputs, learning algorithms help stabilize activation patterns that are functionally significant, reinforcing their role as discrete entities within the system. This process allows connectionist models to implement beliefs that are both functionally discrete and causally significant, bridging the gap between distributed representations and propositional modularity.

In conclusion, the compatibility of functionally discrete beliefs with connectionist models is supported by the ability of these systems to form stable, distinct activation patterns through specialized subnetwork interactions, direct processing toward specific outcomes, and preserve semantic distinctiveness through structural constraints and systematic learning. These features demonstrate that connectionist architectures can feasibly implement beliefs that are both functionally discrete and causally significant, challenging the traditional divide between symbolic and connectionist paradigms in cognitive science.

A.3 GPT-4o Fine-Tuned with Synthesized Instructions

\subsection*{2.2.2. Functionally Discrete Beliefs}

The second aspect of propositional modularity is the notion that beliefs are functionally discrete. This means that beliefs can be individually identified and play distinct roles in cognitive processes. In everyday psychology, we often attribute specific actions to particular beliefs. For instance, we might say that someone went to the store because they believed they needed milk. This implies that the belief about needing milk had a unique causal influence on the decision to go to the store, separate from other beliefs the person might hold.

The question is whether connectionist models can accommodate this idea of functionally discrete beliefs. Can a connectionist system have distinct, identifiable states that correspond to individual beliefs and play specific roles in cognitive processes?

The answer is yes. In a connectionist system, different beliefs can be represented by different patterns of activation across the network. These patterns can be functionally discrete in the sense that each pattern can direct the system's processing toward a particular outcome. For example, one pattern might lead the system to generate a specific response, while another pattern leads to a different response.

Moreover, these patterns can be stable and recurring, meaning that the same pattern can be activated multiple times in different contexts. This

stability allows the patterns to be identified as distinct states of the system, much like individual beliefs in everyday psychology.

However, there is a challenge. In a connectionist network, the same units and connections are involved in representing multiple beliefs. This raises the question of how the network can maintain the distinctness of different patterns without them interfering with each other.

The solution lies in the way connectionist networks are structured and trained. The network's architecture and learning algorithms can impose constraints that ensure different patterns remain distinct. For example, the network might be designed so that certain units are more strongly associated with certain patterns, or the learning algorithm might adjust the weights in a way that preserves the distinctness of different patterns.

In summary, connectionist models can accommodate the idea of functionally discrete beliefs. Through the use of specialized subnetworks and other mechanisms, connectionist systems can form distinct, recurring patterns of activation that play specific roles in cognitive processes. These patterns can be identified as functionally discrete states of the system, much like individual beliefs in everyday psychology.

\subsection*{2.2.3. Beliefs Playing a Causal Role}

The third aspect of propositional modularity is the idea that beliefs play a causal role in cognitive processes. In everyday psychology, we often explain actions by attributing them to specific beliefs. For example, we might say that someone went to the store because they believed they needed milk. This implies that the belief about needing milk had a causal influence on the decision to go to the store.

The question is whether connectionist models can accommodate this idea of beliefs playing a causal role. Can a connectionist system have states that correspond to individual beliefs and have a causal influence on the system's behavior?

The answer is yes. In a connectionist system, different beliefs can be represented by different patterns of activation across the network. These patterns can have a causal influence on the system's behavior in the sense that each pattern can direct the system's processing toward a particular outcome. For example, one pattern might lead the system to generate a specific response, while another pattern leads to a different response.

Moreover, these patterns can be stable and recurring, meaning that the same pattern can be activated multiple times in different contexts. This stability allows the patterns to be identified as distinct states of the system, much like individual beliefs in everyday psychology.

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The solution lies in the way connectionist networks are structured and trained. The network's architecture and learning algorithms can impose constraints that ensure different patterns remain distinct. For example, the network might be designed so that certain units are more strongly associated with certain patterns, or the learning algorithm might adjust the weights in a way that preserves the distinctness of different patterns.

In summary, connectionist models can accommodate the idea of beliefs playing a causal role. Through the use of specialized subnetworks and other mechanisms, connectionist systems can form distinct, recurring patterns of activation that have a causal influence on the system's behavior. These patterns can be identified as functionally discrete states of the system, much like individual beliefs in everyday psychology.

\subsection*{2.2.4. Conclusion}

In conclusion, connectionist models can accommodate the three aspects of propositional modularity: the idea that beliefs are functionally discrete, that they play a causal role, and that they are semantically interpretable. Through the use of specialized subnetworks and other mechanisms, connectionist systems can form distinct, recurring patterns of activation that play specific roles in cognitive processes and have a causal influence on the system's behavior. These patterns can be identified as functionally discrete states of the system, much like individual beliefs in everyday psychology.

A.4 GPT-4o Fine-Tuned with the Same Instruction

\section*{3.2.2. Functionally Discrete Beliefs and Causal Role}

The second aspect of propositional modularity is the concept of functionally discrete beliefs. In everyday psychology, beliefs are treated as distinct entities that can be individually referenced and play a causal role in behavior. For instance, if someone believes that the train leaves at 10:00, this belief might cause them to arrive at the station by 9:45. If they also believe the train leaves at 10:30, they might arrive at 10:15. The belief about the 10:00 departure is seen as a distinct, causally significant entity.

In connectionist models, the question arises: can a network's state be considered a functionally discrete belief? The answer is not straightforward. While a network's state can be seen as a belief, it is not functionally discrete in the same way as beliefs in everyday psychology. A network's state is a complex pattern of activation across many units, and it is not clear how to isolate a single belief from this pattern.

However, it is possible to imagine a connectionist system where certain activation patterns recur in specific contexts, acting as distinct, causally significant states. For example, a network might have a pattern that represents the belief that the train leaves at 10:00, which recurs whenever

this belief is relevant. This pattern could direct the network's processing toward outcomes associated with the belief, such as arriving at the station by 9:45.

In this scenario, the network's state can be seen as a functionally discrete belief. The pattern representing the belief about the 10:00 departure is distinct from other patterns and has a unique causal influence on the network's behavior.

However, there are challenges to this view. Connectionist systems are often highly distributed, with no clear boundaries between different patterns. This raises the question of whether it is possible to have distinct, recurring patterns that are stable enough to be considered functionally discrete beliefs.

The answer depends on the architecture of the connectionist system. In some cases, it may be possible to have specialized subnetworks that interact to produce distinct patterns. These patterns could be stable enough to be considered functionally discrete beliefs.

In summary, it is possible for connectionist models to have functionally discrete beliefs that play a causal role in behavior. However, this depends on the architecture of the system and the ability to form distinct, recurring patterns that are stable enough to be considered beliefs.

\section*{3.2.3. Semantic Interpretation and Distributed Representation}

The third aspect of propositional modularity is the semantic interpretation of beliefs. In everyday psychology, beliefs are seen as having a clear, interpretable meaning. For example, the belief that the train leaves at 10:00 has a clear meaning that can be easily understood.

In connectionist models, the question arises: can a network's state be given a clear, interpretable meaning? The answer is not straightforward. A network's state is a complex pattern of activation across many units, and it is not clear how to assign a single meaning to this pattern.

However, it is possible to imagine a connectionist system where certain activation patterns have a clear, interpretable meaning. For example, a network might have a pattern that represents the belief that the train leaves at 10:00. This pattern could be given a clear, interpretable meaning, similar to the belief in everyday psychology.

In this scenario, the network's state can be seen as having a clear, interpretable meaning. The pattern representing the belief about the 10:00 departure has a clear meaning that can be easily understood.

However, there are challenges to this view. Connectionist systems are often highly distributed, with no clear boundaries between different patterns. This raises the question of whether it is possible to assign a single meaning to a complex, distributed pattern.

The answer depends on the architecture of the connectionist system. In some cases, it may be possible to have specialized subnetworks that interact to produce patterns with a clear, interpretable meaning.

In summary, it is possible for connectionist models to have states with a clear, interpretable meaning. However, this depends on the architecture of the system and the ability to form patterns that can be given a clear meaning.

\section*{3.2.4. Conclusion}

In conclusion, connectionist models can be compatible with the concept of propositional modularity. It is possible for connectionist systems to have functionally discrete beliefs that play a causal role in behavior and have a clear, interpretable meaning. However, this depends on the architecture of the system and the ability to form distinct, recurring patterns that are stable enough to be considered beliefs.

Eigenständigkeitserklärung

Ich versichere hiermit, dass die vorliegende Master-Abschlussarbeit selbständig verfasst wurde und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt wurden.

Unterschrift: